

We are considering the SPDE

$$\begin{cases} \frac{\partial u}{\partial t}(t, x) - Au(t, x) = f(x, u(t, x)) + \sigma(x, u(t, x))\dot{W}(t, x) & x \in G, t > 0 \\ u(0, x) = u_0(x) \end{cases} \quad (0.1)$$

with

- $G = \mathbb{R}_+^d$
- $\rho(x) = \exp(-|x|^2)$
- $H = L_\rho^2(G)$
- $D(A) := H_\rho^2(G) \cap H_{0,p}^1(G)$
- $f : G \times \mathbb{R} \rightarrow \mathbb{R}$ and $\sigma : G \times \mathbb{R} \rightarrow \mathbb{R}$ and satisfy

$$|f(x, u_1) - f(x, u_2)| \leq L|u_1 - u_2|; \quad |\sigma(x, u_1) - \sigma(x, u_2)| \leq K|u_1 - u_2|$$

with L and K independent of x .

- $f(x, 0) \in L^\infty(G)$ and $\sigma(x, 0) \in L^\infty(G)$

We have **Theorem 3** which says that there exists a unique solution, $u^*(t)$, which is defined for all $t \in \mathbb{R}$ and that

$$\sup_{t \in \mathbb{R}} \mathbb{E} \|u^*(t)\|_H < \infty \quad (0.2)$$

and by **Definition 2** that the mild solution for $t \in \mathbb{R}$ is given by

$$u^*(t) = S(t - t_0)u^*(t_0) + \int_{t_0}^t S(t - t_0)f(u^*(s)) \, ds + \int_{t_0}^t S(t - t_0)\sigma(u^*(s))dW(s) \quad -\infty < t_0 < t < \infty \quad (0.3)$$

and in addition, in the linear case $du = (Au + \alpha(t))dt + \phi(t)dW(t)$ we have that

$$u^*(t) = \int_{-\infty}^t S(t - s)\alpha(s) \, ds + \int_{-\infty}^t S(t - s)\phi(s) \, dW(s) \quad (0.4)$$

and in the nonlinear we have that $u^*(t)$ is a limit of solutions of the form (0.4).

My question is the following:

In (0.1) we see that $u(0, x) = u_0(x)$ but in the solution (0.3), it doesn't seem obvious to me that in the general case

$$\begin{aligned} u^*(0) &= S(0 - t_0)u^*(t_0) + \int_{t_0}^0 S(0 - t_0)f(u^*(s)) \, ds + \int_{t_0}^0 S(0 - t_0)\sigma(u^*(s))dW(s) \\ &= u(0, x) \end{aligned}$$

or even in the linear case

$$\begin{aligned} u^*(0) &= \int_{-\infty}^0 S(0 - s)\alpha(s) \, ds + \int_{-\infty}^0 S(0 - s)\phi(s) \, dW(s) \\ &= u(0, x) \end{aligned}$$

I don't see where the initial information $u(0, x) = u_0(x)$ is used when proving that (0.4) satisfies the conditions in **Definition 2** for the mild solution. It seems like the initial condition didn't play any role in finding the solution for $t \in \mathbb{R}$. I was hoping you could clear up my confusion. Thank you very much in advance!